ECDLP follow-ups: factor-base sweep, pencil pilot and Kummer gate

Reproducible experiment report

21 October 2026

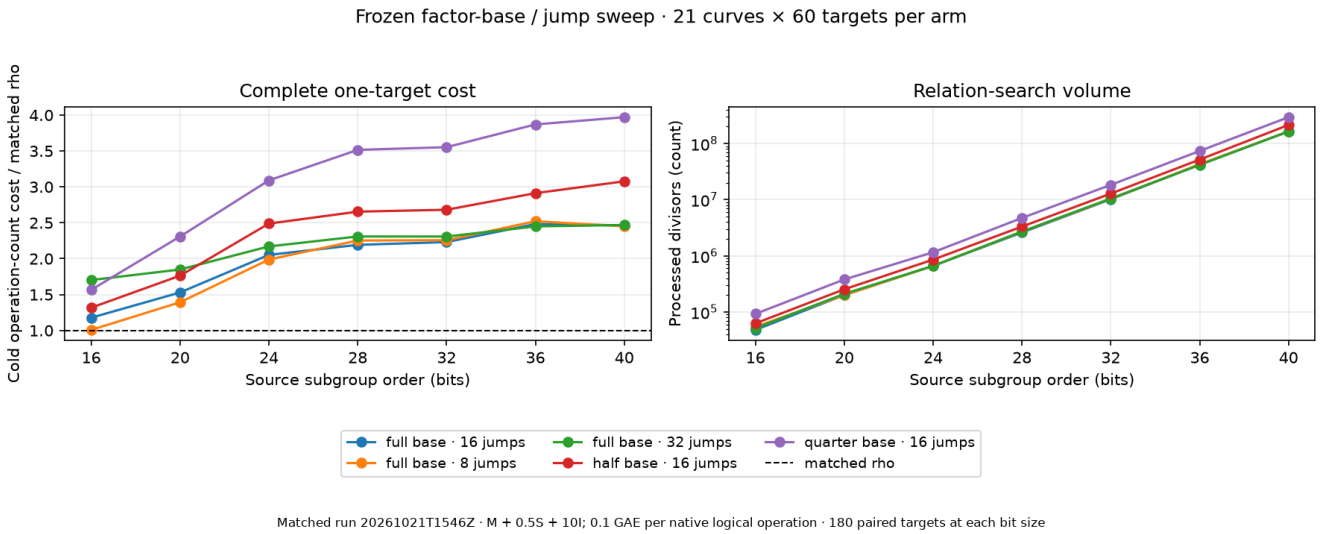
Scope. The frozen protocol covers four leads on 21 exact-order Scholten models, 60 separate targets per model. This report contains completed factor-base and jump experiments, a 240-target pencil pilot, and an exact Kummer applicability gate. The complete inversion-free law and full-cohort pencil collector remain open. All costs are cold one-target operation-count prices, not wall-speed claims.

Factor-base and jump sweep

Five arms solved and source-verified all 1,260 targets each, with zero failures or cap exhaustions. All 6,300 same-process rho controls were also source-verified. All 1,260 incumbent count and phase records replay exactly. At $M + 0.5S + 10I$ and 0.1 source-addition equivalents (GAE) per declared native logical operation, matched negation rho totals 1,088,802,370 GAE.

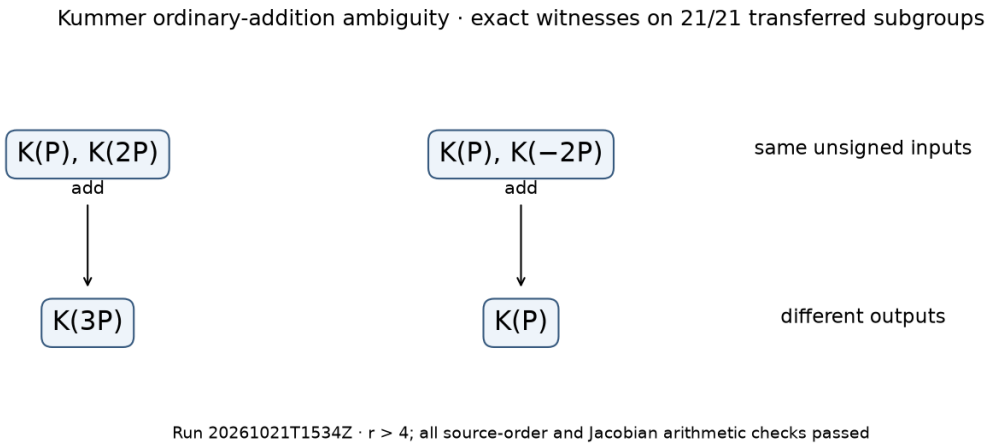
Arm	Divisors processed	Total GAE	/ external	/ rho
Full base, 16 jumps	219,581,773	2,666,291,894	1.000000	2.448830
Full base, 8 jumps	220,092,171	2,664,487,778	0.999323	2.447173
Full base, 32 jumps	219,864,588	2,671,135,032	1.001816	2.453278
Half base, 16 jumps	282,983,105	3,285,713,695	1.232316	3.017732
Quarter base, 16 jumps	392,583,830	4,267,226,320	1.600435	3.919193

The half and quarter bases rejected 35,385,164 and 110,402,348 split relations, respectively, when neither root was in the fixed target-blind base; all rejected candidates were charged. None of the five arms is below rho in the 36 pooled pricing cells. The 8-jump arm is below the 16-jump arm in 15 of 36 cells. Across bit sizes it is below at 16, 20, 24 and 40 bits, and above at 28, 32 and 36 bits. One seed was used per target; no uncertainty interval or isolated wall-time result was measured.



Kummer ordinary-addition gate

Set $P = \tau(G)$ and $Q = 2P$. The unsigned quotient gives $K(Q) = K(-Q)$, but $K(P+Q) = K(3P)$ and $K(P-Q) = K(P)$ differ for order $r > 4$. The source-order, Jacobian-arithmetic and distinct-output checks passed on 21/21 exact-order models. This rules out binary ordinary addition from two unsigned Kummer inputs for the current arbitrary-jump walk. A differential walk with a known difference and sign/lift recovery has not been implemented or priced.

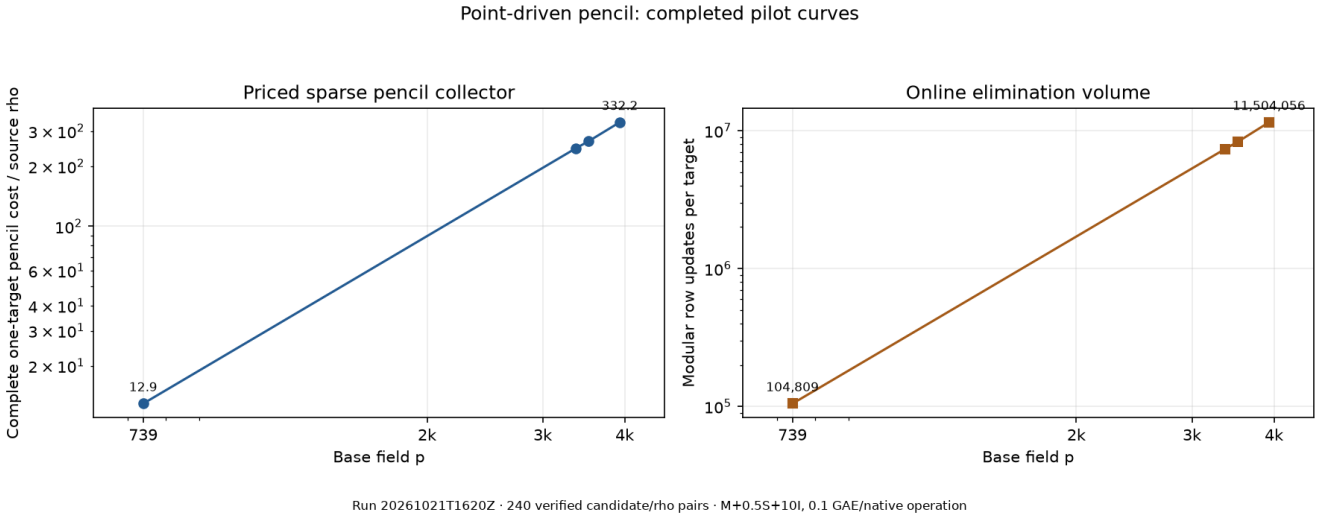


Point-driven pencil pilot

Four original models, $p = 739, 3371, 3527, 3931$, each ran 60 separate targets. All 240 candidate and same-process rho scalars passed independent source multiplication. The replay matched every non-wall field in the four raw files. The direct F739 port also reproduced all 61 prior records after removing wall fields. Full measured setup cost is charged to each cold one-target solve.

p	Verified	Relations	Modular row updates	Pencil/rho
739	60/60	21,550	6,288,519	12.931178
3,371	60/60	98,666	442,470,426	246.117110
3,527	60/60	103,256	499,795,794	267.374941
3,931	60/60	114,989	690,243,338	332.187349

The price is $M + 0.5S + 10I$ plus 0.1 GAE per native logical operation; the independent validator recomputed 12 headline values from raw rows and matched fresh rho counters to frozen controls. The 36-cell price grid and per-target JSONL are in `results/pencil-pilot/`. Upper-size runs are still open.



Open requirements and evidence

The published complete inversion-free genus-2 formulas have not been implemented or run on the 1,260 targets. The point-driven pencil collector has 240 new complete target solves but no full-cohort result here. A differential Kummer collector remains unimplemented. These are open requirements, not negative end-to-end verdicts.

The machine record consists of 35 raw matched JSONL arm files, 6,300 paired candidate/rho target rows, the full 36-cell price grid, summary JSON, 21 Kummer witnesses, four complete pencil JSONL files with a 36-cell-per-model price grid, and a checksum manifest under `research/four_followups_20261021/`. The native analyzer pairs workload hashes, seeds, caps and source scalars; the full-base arm reproduces the frozen external-law counts and phases. A separate native validator recomputed 15 headline quantities from raw rows; an archive-input rerun was compared byte-for-byte across all 35 files. The plotting script only renders those recorded values. See https://github.com/aburan28/crypto/pull/2304 for the review branch and https://eprint.iacr.org/2024/623.pdf for the separate complete-law path.